

Auditing Heterotic Flux Compactifications on Iwasawa and Lens–Nil Geometries: Bundle Obstructions and a $q = 79$ Fu–Yau Completion Contract

Peter Nero

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Abstract

We determine which parts of two proposed heterotic flux constructions survive the full Hull–Strominger proof obligations. The diagonal invariant Hermitian structure on the Iwasawa manifold is balanced, and its torsion and dH are computed exactly. The proposed bundle, however, does not exist as claimed. One of its alleged first-Chern forms is not closed; the monad maps were not constructed as global sections; the printed Dolbeault operator is not integrable; and a connection on the stated trivial smooth rank-three carrier cannot have third Chern number six. Within the declared one-entry repair ansatz, the Maurer–Cartan equation has a unique signed repair, but the repaired family is a single complex-gauge orbit with a non-scalar holomorphic commutant, so it cannot supply the claimed stable bundle. The old anomaly, three-generation, and normalized-Yukawa conclusions therefore do not follow. The Lens–Nil model is a balanced real $SU(3)$ -structure comparison model, but its almost-complex structure is non-integrable and hence it is not a Hull–Strominger compactification. We finish by separating the exact $q = 79$ finite and rank-two Hermitian–Yang–Mills evidence from the still-open physical rank-three problem, and state a typed completion contract for a future Fu–Yau endpoint.

Version 5 Revision Note

Supersedes. Version 4, which remains the released edition. **Reason and resolution.** Incorporates the new local $q79$ response certificates and the correction separating a fixed-fiber Picard test from the global BHT obstruction. **Retained results.** The Iwasawa audit and typed compactification contract are unchanged. **Boundary.** The new certificates do not construct the selected common physical endpoint.

Contents

1	Introduction	2
1.1	What is proved and what is not	3
2	The proof obligations for a heterotic solution	3
3	The surviving Iwasawa geometry	4
4	Why the proposed rank-three bundle fails	5
4.1	The monad data were not global bundle data	5
4.2	The printed Dolbeault operator is not integrable	5

5	Consequences for anomaly cancellation and Yukawa claims	6
5.1	The Bianchi identity	6
5.2	Generation number and Yukawa coupling	7
6	Why Lens–Nil is not a Hull–Strominger solution	7
7	Baseline q79 evidence and its exact scope	7
7.1	Available ingredients	8
7.2	Why the Fu–Yau direction is different	8
7.3	September update: local observability and the global obstruction	9
8	A typed completion theorem	9
9	Interpretation inside MTT	10
10	Discussion	10
11	Reproducibility	11
Revision note: Version 4		11
12	Conclusion	12

1 Introduction

The Hull–Strominger system is attractive because geometry, gauge data, torsion, and anomaly cancellation must all hold on the same compact complex threefold. That coupling also makes proposed solutions unusually easy to overstate. A balanced metric is not yet a heterotic vacuum; a formal matrix of $(0,1)$ -forms is not yet a holomorphic bundle; a cohomological Chern-class identity is not yet the differential Bianchi identity; and a topological index is not yet a computed four-dimensional Yukawa coupling.

An earlier version of this paper presented explicit solutions on the Iwasawa manifold and on a Lens–Nil geometry. The present version re-executes those claims against the mathematical type of every object. The result is mixed but useful:

1. the local Iwasawa $SU(3)$ -structure and torsion calculation survives;
2. the proposed Iwasawa rank-three bundle, anomaly match, generation count, and Yukawa normalization do not survive;
3. Lens–Nil survives as an auxiliary real $SU(3)$ -structure model, but not as a Hull–Strominger background;
4. the selected $q = 79$ branch remains a plausible Fu–Yau-oriented research direction, with exact finite and separate rank-two HYM results, but its physical visible–hidden bundle endpoint is still open.

This is not a no-go theorem for all heterotic solutions on the Iwasawa manifold. Invariant solutions on complex Lie groups and nilmanifolds are known under carefully specified choices of tangent and gauge connections [4, 3]. The result here is narrower: the particular bundle and coefficient chain printed in the earlier paper does not establish such a solution.

1.1 What is proved and what is not

Object	Status here	Meaning
Iwasawa diagonal $SU(3)$ structure	proved	Complex, balanced, non-Kähler; H and dH are explicit.
Old Iwasawa monad	withdrawn	Its Chern data and global maps were not defined.
Old Dolbeault matrix	disproved	Its Maurer–Cartan residual is nonzero.
Minimal signed repair	proved but rejected	Integrable, yet gauge-redundant and non-simple; not a stable visible bundle.
Old Bianchi and Yukawa chain	withdrawn	It depends on the invalid bundle and on further unproved inputs.
Lens–Nil model	auxiliary only	Balanced as a real $SU(3)$ structure, but non-integrable.
$q = 79$ Fu–Yau endpoint	open	Finite/topological and rank-two ingredients exist; the common physical rank-three Hull–Strominger tuple has not been constructed.

2 The proof obligations for a heterotic solution

Let X be a compact complex threefold with a nowhere-vanishing holomorphic $(3,0)$ -form Ω , Hermitian form ω , dilaton Φ , gauge bundle V , and unitary connections A on V and ∇ on TX . In one standard first-order convention, the Hull–Strominger equations include

$$d(\|\Omega\|_\omega \omega^2) = 0, \quad (1)$$

$$F_A^{0,2} = 0, \quad F_A \wedge \omega^2 = 0, \quad (2)$$

$$H = i(\bar{\partial} - \partial)\omega, \quad dH = \frac{\alpha'}{4} (\text{tr} R_\nabla \wedge R_\nabla - \text{tr} F_A \wedge F_A). \quad (3)$$

Trace normalization and the choice of ∇ must be fixed consistently. If the supersymmetry equations are also required to imply the first-order equations of motion, the tangent connection must obey the corresponding instanton condition. These choices cannot be changed midway through a coefficient comparison [11, 7, 2].

In the mathematical normalization one identifies $\|\Omega\|_\omega$ with the dilaton factor up to convention. For constant dilaton and constant $\|\Omega\|_\omega$, Equation (1) reduces to $d(\omega^2) = 0$. This balanced condition is one row of the system, not a replacement for the other rows.

Definition 2.1 (Complete first-order certificate). A complete certificate for a background in the convention above contains:

- (i) an integrable complex threefold and global Ω ;
- (ii) a positive Hermitian form satisfying the conformally balanced equation;
- (iii) actual holomorphic gauge bundles and HYM connections;
- (iv) a specified tangent connection and its curvature;
- (v) equality of the differential four-forms in Equation (3);
- (vi) the required integral or differential-cohomological flux data.

Topological equality $c_2(TX) = c_2(V)$ can be necessary, but it does not determine the chosen curvature representatives. Likewise, the Li–Yau theorem supplies an HYM connection after the relevant stability hypotheses are proved; it does not turn a partial search for invariant sections into a stability proof [8]. Established constructions make these dependencies explicit, for example the Fu–Yau solutions and stable-bundle perturbative constructions [6, 1].

3 The surviving Iwasawa geometry

Let $X = \Gamma \backslash H_3(\mathbb{C})$ be an Iwasawa manifold with global left-invariant $(1, 0)$ -forms

$$d\omega^1 = d\omega^2 = 0, \quad d\omega^3 = \omega^1 \wedge \omega^2. \quad (4)$$

For $r_j > 0$, set

$$\omega = \frac{i}{2} \sum_{j=1}^3 r_j^2 \omega^j \wedge \bar{\omega}^j, \quad \Omega = \omega^1 \wedge \omega^2 \wedge \omega^3. \quad (5)$$

Proposition 3.1 (Exact invariant geometry). *The structure in Equation (5) is complex, balanced, and non-Kähler. With $H = i(\bar{\partial} - \partial)\omega$,*

$$H = \frac{r_3^2}{2} (\omega^3 \wedge \bar{\omega}^1 \wedge \bar{\omega}^2 + \omega^1 \wedge \omega^2 \wedge \bar{\omega}^3), \quad (6)$$

$$dH = r_3^2 \omega^1 \wedge \bar{\omega}^1 \wedge \omega^2 \wedge \bar{\omega}^2. \quad (7)$$

Proof. The structure equations have no $(0, 2)$ component, so the declared almost-complex structure is integrable. Also $d\Omega = 0$, because the only possible term contains $\omega^1 \wedge \omega^2 \wedge \omega^1 \wedge \omega^2$. The only nonclosed summand of ω is the $j = 3$ summand. When $d\omega$ is wedged with either of the remaining diagonal summands, a repeated ω^1 or ω^2 occurs. Hence $d(\omega^2) = 2d\omega \wedge \omega = 0$. Since $d\omega \neq 0$, the metric is not Kähler. Splitting Equation (4) into types gives

$$\partial\omega = \frac{ir_3^2}{2} \omega^1 \wedge \omega^2 \wedge \bar{\omega}^3, \quad \bar{\partial}\omega = \frac{ir_3^2}{2} \omega^3 \wedge \bar{\omega}^1 \wedge \bar{\omega}^2,$$

which yields Equations (6) and (7). □

Define the invariant diagonal $(1, 1)$ -forms

$$a = \frac{i}{2} \omega^1 \wedge \bar{\omega}^1, \quad b = \frac{i}{2} \omega^2 \wedge \bar{\omega}^2, \quad c = \frac{i}{2} \omega^3 \wedge \bar{\omega}^3. \quad (8)$$

Then Equation (7) is equivalently

$$dH = -4r_3^2 a \wedge b. \quad (9)$$

The earlier construction treated all three forms in Equation (8) as first-Chern representatives. That is the first decisive type error.

Proposition 3.2 (Closed-form obstruction). *Within the span of a, b, c , a form $xa + yb + zc$ is closed if and only if $z = 0$. In particular, c cannot represent the first Chern class of a line bundle.*

Proof. The first two forms are closed, while

$$dc = \frac{i}{2} (\omega^1 \wedge \omega^2 \wedge \bar{\omega}^3 - \omega^3 \wedge \bar{\omega}^1 \wedge \bar{\omega}^2) \neq 0. \quad (10)$$

The $(2, 1)$ and $(1, 2)$ summands in Equation (10) are linearly independent, so $zdc = 0$ exactly when $z = 0$. A Chern–Weil representative of c_1 is closed; therefore c is ineligible. Closedness is only the first gate: integrality must still be checked by periods on the chosen lattice. □

4 Why the proposed rank-three bundle fails

4.1 The monad data were not global bundle data

For a genuine monad

$$0 \longrightarrow K_1 \xrightarrow{f} \bigoplus_i L_i \xrightarrow{g} K_2 \longrightarrow 0,$$

the L_i, K_j must first exist as line bundles. Every entry of f and g must then be a global holomorphic section of the appropriate Hom bundle, for example

$$g_i \in H^0(X, \text{Hom}(L_i, K_2)) = H^0(X, K_2 \otimes L_i^{-1}).$$

A “constant matrix in a left-invariant frame” is not automatically such a section when its source and target line bundles are nonisomorphic.

The former monad used combinations containing c as first-Chern data. Proposition 3.2 prevents those line bundles from being defined with the advertised curvature. No global Hom sections, exactness loci, or locally free cohomology sheaf were supplied independently. Consequently the formal Chern-character arithmetic did not compute the Chern classes of an actual rank-three bundle.

The old stability argument had a second logical gap. The statement $H^0(X, E) = 0$, even if proved, excludes a map $\mathcal{O}_X \rightarrow E$; it does not exclude every line bundle L of nonnegative slope from mapping into E . Nor does checking invariant candidates exhaust all coherent subsheaves. Slope stability requires every proper torsion-free subsheaf to have smaller slope. The Li–Yau theorem becomes applicable only after that global condition is established.

4.2 The printed Dolbeault operator is not integrable

Write a left-invariant candidate as

$$\bar{\partial}_E = \bar{\partial} + A, \quad A = B_1 \bar{\omega}^1 + B_2 \bar{\omega}^2 + B_3 \bar{\omega}^3. \quad (11)$$

Because $\bar{\partial} \bar{\omega}^3 = \bar{\omega}^1 \wedge \bar{\omega}^2$, the Maurer–Cartan equation is

$$\bar{\partial}_E^2 = 0 \iff B_3 + [B_1, B_2] = 0. \quad (12)$$

The earlier paper used

$$B_1 = \sqrt{\mu} E_{13}, \quad B_2 = -\sqrt{\mu} E_{31}, \quad B_3 = \mu E_{12}, \quad \mu > 0. \quad (13)$$

Substitution gives

$$B_3 + [B_1, B_2] = \mu(E_{12} - E_{11} + E_{33}) \neq 0.$$

Thus the printed operator does not define a holomorphic structure.

Theorem 4.1 (Minimal signed repair and its obstruction). *Within the declared one-entry signed repair in which the B_2 arrow is moved to the composable Heisenberg arrow, Equation (12) forces*

$$B_1 = \sqrt{\mu} E_{13}, \quad B_2 = -\sqrt{\mu} E_{32}, \quad B_3 = \mu E_{12}. \quad (14)$$

This family is one $SL(3, \mathbb{C})$ complex-gauge orbit. Its holomorphic commutant has complex dimension two, so it is not simple and cannot be the claimed stable rank-three bundle.

Proof. Since $E_{13}E_{32} = E_{12}$ and $E_{32}E_{13} = 0$, $[B_1, B_2] = -\mu E_{12}$, proving integrability. For

$$G_\mu = \text{diag}(\sqrt{\mu}, \mu^{-1/2}, 1) \in SL(3, \mathbb{C}),$$

one has $B_j(\mu) = G_\mu B_j(1) G_\mu^{-1}$ for all three matrices. Hence μ labels complex gauge, not distinct holomorphic data. The matrices commuting with E_{13} , E_{32} , and E_{12} are exactly

$$\{ \lambda I + \nu E_{12} : \lambda, \nu \in \mathbb{C} \}.$$

The non-scalar element E_{12} is therefore a holomorphic endomorphism. A slope-stable holomorphic vector bundle is simple, so the repaired object cannot satisfy the claimed stability conclusion. $\square$

Corollary 4.2 (Topological obstruction on the stated carrier). *If Equation (11) is a connection on the stated smooth carrier $E_{\text{sm}} = X \times \mathbb{C}^3$, then*

$$c_1(E_{\text{sm}}) = c_2(E_{\text{sm}}) = c_3(E_{\text{sm}}) = 0.$$

In particular, it cannot have $\int_X c_3 = 6$.

Proof. The total Chern class of a trivial smooth bundle is one. Changing a connection changes a Chern–Weil representative by an exact transgression form, not the underlying characteristic class. A nonzero local six-form expression cannot override this topological fact. $\square$

5 Consequences for anomaly cancellation and Yukawa claims

The preceding obstruction propagates to every downstream claim that used the old bundle.

5.1 The Bianchi identity

Complex parallelizability makes the holomorphic tangent bundle of the Iwasawa manifold trivial, so $c_2(TX) = 0$. This gives a cohomological simplification, not the differential identity in Equation (3). That identity still requires specified connections and their actual curvature four-forms.

The old hidden abelian flux also does not pass the stated HYM check. Its nonzero component was proportional to $a + 2b$. For the positive diagonal metric,

$$\Lambda_\omega(a + 2b) > 0,$$

so it is not primitive. Placing the negative of the same form along an independent Cartan generator does not cancel a Lie-algebra-valued primitivity condition; each independent component must vanish. In addition, integer coefficients in an invariant coframe do not prove flux quantization until the period lattice is computed.

Because the visible connection is invalid, the hidden flux is not HYM as written, and the tangent-curvature convention was not independently certified, the former coefficient equation for r_3 is withdrawn. A rebuilt solution must recompute all three curvature terms from one declared set of connections.

5.2 Generation number and Yukawa coupling

For a genuine special-unitary bundle on a compact complex threefold, an index formula can relate net chirality to $\frac{1}{2} \int_X c_3(V)$ under the standard hypotheses. Corollary 4.2 shows that the old carrier does not provide the advertised value. The three-generation conclusion therefore does not follow from this Iwasawa construction.

A Yukawa coupling requires considerably more data:

1. a valid holomorphic bundle and its relevant cohomology groups;
2. normalized harmonic representatives for the matter modes;
3. the bundle contraction or invariant tensor;
4. evaluation of the corresponding overlap integral.

Orthonormality of three representatives does not imply that their wedge product equals $\bar{\Omega}$, even up to phase. The old assertion $\lambda_{123} = 1$ and its rank-one mass interpretation are therefore withdrawn. This does not conflict with separate MTT finite/profile Yukawa results; it says only that this compactification did not derive them.

6 Why Lens–Nil is not a Hull–Strominger solution

A real six-manifold can carry an $SU(3)$ structure (ω, Ω) even when its induced almost-complex structure is not integrable. Conditions such as $d(\omega^2) = 0$ still make sense in that setting. The Hull–Strominger system, however, uses a complex threefold, holomorphic bundles, and the Dolbeault operators $\partial, \bar{\partial}$ with $\bar{\partial}^2 = 0$.

The former Lens–Nil section explicitly found $d\Omega \neq 0$ and described the structure as non-integrable. It then used $H = i(\bar{\partial} - \partial)\omega$, holomorphic bundle language, and Hull–Strominger anomaly equations as if integrability had been established. Those steps are incompatible.

Proposition 6.1 (Correct status of the Lens–Nil construction). *The balanced Lens–Nil ansatz from the earlier paper is an auxiliary real $SU(3)$ -structure model. It is not a solution of the Hull–Strominger system in the complex-geometric sense used in Section 2.*

Proof. Integrability is a prerequisite for the Dolbeault complex and for holomorphic gauge bundles. The ansatz fails that prerequisite by its own $d\Omega \neq 0$ calculation. Balancedness alone cannot restore it. $\square$

This conclusion does not say that Lens or Nil geometry is useless in MTT. It can remain an auxiliary spectral, transport, or rank-filter model. What must be retired is the inference from that auxiliary role to a literal physical Hull–Strominger compactification. The literal $L(3, 1) \times \text{Nil}_3$ model is also not the selected $q = 79$ Fu–Yau candidate; distinct topology cannot be erased by assigning the same interpretive labels.

7 Baseline q79 evidence and its exact scope

The failure of the old Iwasawa bundle does not return the broader MTT program to zero. It changes which results may be composed.

7.1 Available ingredients

The current research ledger contains the following separate results:

1. an exact finite arithmetic selection of the $q = 79$ branch;
2. smooth non-pullback rank-three topological candidates with $\int c_3 = \pm 6$, giving index ± 3 at the topological tier;
3. an exact finite rank-two Cech witness with 81 transition entries and 729 triple-overlap checks;
4. a certified finite HYM approximation with residual 8.208×10^{-13} , coercivity lower bound 26.0187, and residual-to-gap indicator 3.155×10^{-14} ;
5. a Wiener-algebra contraction certificate for that selected rank-two problem, with contraction constant $Z = 0.38508 < 1$ and verified ball bound $Y + Zr = 0.00932703 < 0.01$, yielding existence and local uniqueness in the declared neighborhood.

The machine-readable packets and verifiers are curated in the MTT Results Reproduction repository [9]. The topological rank-three/index statement is also discussed separately in Ref. [10].

These are meaningful results, but their ranks and domains matter. The rank-two Cech/HYM witness is not the missing rank-three visible bundle. The smooth rank-three $c_3 = \pm 6$ candidate is not yet a holomorphic stable bundle with an HYM connection. Neither result by itself supplies a common visible–hidden pair, the differential Bianchi representative, or a worldsheet theory.

7.2 Why the Fu–Yau direction is different

Fu–Yau geometry belongs to an established class of complex non-Kähler heterotic constructions: torus bundles over a K3-type base can support balanced metrics and solutions of the anomaly equation under explicit topological and analytic hypotheses [6, 5]. This makes it a mathematically appropriate target for the selected MTT branch. It does not make every $q = 79$ topological carrier a Fu–Yau solution automatically.

The current branch must still produce the following objects on one and the same complex threefold X_{79} :

Gate	Required object	Current state
G1	Global complex X_{79} , holomorphic volume form, balanced Fu–Yau metric	candidate/partial
V1	Physical rank-three visible holomorphic bundle with $\int c_3 = \pm 6$	open
V2	Compatible hidden holomorphic bundle or sheaf data	open
H1	Visible, hidden, and tangent instantons in one metric chamber	rank-two evidence only
A1	Differential Green–Schwarz identity with fixed traces and connections	open
Q1	Global gerbe/flux quantization and patching	open
Y1	Matter cohomology, normalized overlaps, and physical Yukawa map	open
W1	Exact heterotic worldsheet and modular/factorization data	open downstream

7.3 September update: local observability and the global obstruction

The baseline above is not an exhaustive inventory of the later q79 research. There is now a full selected 164×164 local Gauss–Manin coefficient source with a certified correlated lift on $0 \leq u \leq 2^{-96}$. Separately, the completed same-source affine Deligne test proves $\beta_C(B89) \neq 0$. Thus B89 is rejected from the zero locus; neither the small local interval nor that rejection determines the global answer for every other cover. These results narrow an actual construction problem without changing the ranks or carriers of the baseline HYM theorems.

One correction is particularly important for subsequent searches. The non-torsion calculation at an initial Picard point on one genus-82 fiber does not compute the full global BHT obstruction. CBF T69 retracts the attempted global exclusion of ranks 1 through 1449. The fixed-fiber calculation survives, while the global rank-one and rank-two alternatives remain undecided. The valid implication that an actual rank- r twisted spectral object requires $r\beta_C = 0$ must be applied to that global class, not to a substitute evaluated at one fiber.

The newer response certificates explain why comparing several fibers is useful. CBF T70 finds rank 70 for one coefficient evaluation of a 122-dimensional projective tangent space, leaving a 52-dimensional kernel. CBF T71–T72 instead recover rank 122 from three evaluations. T72 proves the characteristic-zero statement at the three physical midpoints and preserves the coefficient rank on independent boxes $|s_i - 1/2| \leq 2^{-32}$, with certified Neumann defect below 0.199.

CBF T73 supplies the canonical-dual response at those midpoints. K3 adjunction identifies the quotient $H^0(\mathcal{O}(9H))/\langle F \rangle$ with an 82-dimensional canonical space. The three bilinear blocks are invertible and have total rank 246; postcomposition preserves the coefficient rank 122. This last bilinear invertibility is a midpoint result, not yet a certificate over the whole three-panel boxes. In particular, none of these coefficient ranks is silently identified with the derivative of the nonlinear global Picard/BHT restriction map.

All statements in this update are imported from the frozen curated result manifest at commit f141a20e, including the records CBF T69 (corrected global-rank scope), CBF T72 (three-evaluation frame), and CBF T73 (canonical-dual response). The certificates and their original proofs remain the theorem owners. The next bridge is to certify the finite-trace bilinear over all three panels, complete global transport and the integral readout, and evaluate the actual global obstruction. The common HYM/Hull–Strominger endpoint and its physical overlaps still require the remaining rows below.

8 A typed completion theorem

The useful way forward is to make completion compositional without allowing objects from different carriers or ranks to be silently merged.

Theorem 8.1 (Hull–Strominger assembly contract). *Fix one compact complex threefold X , one Hermitian metric ω , one holomorphic volume form Ω , one dilaton Φ , holomorphic bundles $V_{\text{vis}}, V_{\text{hid}}$, unitary connections $A_{\text{vis}}, A_{\text{hid}}$, and a specified unitary tangent connection ∇ . Suppose:*

- (i) Ω is nowhere vanishing and $d(\|\Omega\|_\omega \omega^2) = 0$;
- (ii) both gauge curvatures have type $(1, 1)$ and are primitive;
- (iii) R_∇ obeys the tangent-instanton condition required by the chosen first-order convention;
- (iv) with $H = i(\bar{\partial} - \partial)\omega$, the differential identity

$$dH = \frac{\alpha'}{4} \left(\text{tr} R_\nabla^2 - \text{tr} F_{\text{vis}}^2 - \text{tr} F_{\text{hid}}^2 \right)$$

holds with one fixed trace normalization;

- (v) *the resulting H and Chern–Simons data define the required global differential-cohomological flux class.*

Then these data form a first-order heterotic Hull–Strominger background in the declared convention.

Proof. Each hypothesis is exactly one typed equation or global patching requirement from Section 2. Because every term is defined on the same carrier with fixed connections and traces, the equations can be assembled without changing representatives or importing a result from a different rank. The conclusion is therefore the conjunction of the verified rows. $\square$

Remark 8.2 (What the theorem does not add). Theorem 8.1 is an assembly theorem, not an existence theorem. It prevents a common logical error: satisfying each row somewhere is not the same as satisfying all rows on one selected object. The open MTT task is to construct the tuple satisfying its hypotheses on X_{79} .

Three-family physics requires an additional layer. One must prove the relevant index on V_{vis} , identify the massless cohomology groups, choose the physical embedding and normalization, and compute the overlap map. None of those conclusions is supplied merely by Theorem 8.1.

9 Interpretation inside MTT

The MTT language of selection and admissibility remains useful if it is kept typed. It can organize a search over candidate complex manifolds, bundles, connections, and finite operators. It cannot convert a failed Maurer–Cartan equation into a holomorphic bundle or replace an analytic existence theorem.

The corrected relation among the recurring geometric labels is:

$$\text{local Circle–Lens–Nil filtration} \not\equiv L(3, 1) \times \text{Nil}_3 \not\equiv X_{79}.$$

Circle, lens, and nil may label phase/holonomy, finite transport, and anchoring or shear sectors after projection. The shared circle is common line-bundle phase data counted once; it is not automatically a compact time coordinate. A connection-preserving intertwiner is required before the local filtration can be identified with vertical operators on the global $q = 79$ geometry.

This distinction explains why the rank-two HYM theorem is valuable but not decisive for the physical compactification. It demonstrates that the selected analytic machinery can close a nontrivial HYM problem. It does not change the bundle rank, Chern class, or carrier on which that theorem was proved.

10 Discussion

The main positive result is methodological as well as algebraic. The Iwasawa calculation supplies a clean example of how far local geometry can progress before the global bundle layer becomes decisive. The exact repair in Theorem 4.1 is also informative: integrability can be restored without producing the desired physics. Gauge redundancy and simplicity are independent gates.

The negative conclusions are deliberately local to the constructions audited here:

- they do not rule out other invariant Iwasawa solutions with properly chosen instantons;

- they do not rule out generalized or nonsupersymmetric uses of a non-integrable Lens–Nil $SU(3)$ structure;
- they do not invalidate the exact $q = 79$ finite theorem, the topological rank-three candidate, or the separate rank-two HYM theorem;
- they do prevent those separate results from being advertised as one completed physical compactification.

The most direct research target is now the physical visible bundle gate V1. A successful construction should begin with an actual holomorphic rank-three bundle on the selected Fu–Yau carrier, prove its stability in the same balanced chamber used by the geometry, and only then solve the hidden and Bianchi rows. Reusing the rank-two Wiener-contraction architecture is reasonable; reusing its conclusion without a rank-three operator is not.

11 Reproducibility

The exact finite-dimensional statements behind Theorem 4.1 are checked by the standard-library script `verify_bundle_audit.py` distributed with this paper. It verifies:

1. the nonzero Maurer–Cartan residual of the printed matrix;
2. the vanishing residual of the repaired matrix;
3. the three $SL(3, \mathbb{C})$ conjugation identities as Laurent-polynomial identities;
4. the two-dimensional commutant by exact rational row reduction.

The wider q79, Cech, finite-HYM, and Wiener-contraction artifacts are provided with provenance hashes and their own verifiers in Ref. [9]. Numerical execution does not alter the scope boundaries stated in Section 7.

Revision note: Version 4

Supersedes

Version 4 supersedes Version 3 and its claim of two explicit heterotic compactifications.

Reason

The earlier paper combined a correct local Iwasawa calculation with nonclosed Chern data, an unconstructed monad, a nonintegrable Dolbeault operator, an impossible Chern number on a trivial carrier, and a non-integrable Lens–Nil model.

Resolution

This version proves the closed-form and Maurer–Cartan obstructions, constructs and diagnoses the unique minimal signed repair, withdraws the dependent Bianchi, generation, and Yukawa conclusions, and replaces the claimed solutions by the typed q79 Fu–Yau completion contract.

Retained content

The verified Iwasawa complex and balanced structure, the corrected torsion and dH , the auxiliary role of Lens and Nil filtrations, and Fu–Yau geometry as the strongest current q79 heterotic direction are retained.

Remaining boundary

The program still needs one common q79 rank-three visible-hidden Hull-Strominger tuple satisfying the holomorphic, HYM, differential Bianchi, flux, and matter-overlap gates of Theorem 8.1.

12 Conclusion

The earlier paper did not construct the two claimed heterotic compactifications. What survives is a correct Iwasawa balanced-geometry calculation, an exact diagnosis of the failed bundle, and a sharper route forward. The selected $q = 79$ program has nontrivial finite, topological, and rank-two analytic evidence, but a physical Hull-Strominger compactification still requires one common rank-three visible-hidden tuple satisfying every gate in Theorem 8.1. That is the precise frontier.

Computational Evidence and Reproducibility

The q79 arithmetic theorem and audit, literal finite rank-two Cech witness, and rank-two Wiener-contraction certificate are used directly at their declared finite, topological, or rank-two analytic tiers. They do not construct the missing physical rank-three visible-hidden bundle, differential Green-Schwarz representative, flux gerbe, or worldsheet theory. The paper’s SU3 repair obstruction is checked by its own exact local verifier and is not relabeled as a promoted q79 HYM endpoint.

The referenced rows are frozen to the curated results repository state identified below. Hashes are grouped in eight-character blocks for line breaking.

Repository: <https://github.com/PeterNero/mtt-results-repro>
Commit: 31247ebb 5c22f3fb b5443024 365433c6 ee0bff4a
Manifest: release/result_manifest.json
Manifest SHA-256:
fb399689 60b00584 631dbf53 1a708e18
ef928d6b 6d935119 c185d7f6 32b1e7cd

Tier labels are quoted verbatim from that manifest. A row used directly supports only the specific computational statement identified above; a corpus-state cross-check does not prove this paper’s local theorems; and an open row is evidence of an unresolved obligation, never of closure.

Rows used directly in this paper.

- A19/hym_wiener_contraction (*numeric certified*).
Weighted-theta Fourier-tail and Wiener contraction certificate.
- A07/literal_cech_witness (*derived exact*).
Literal 81-entry, 729-cocycle finite Cech witness.
- A11/q79_exact_audit (*derived exact*).
Executable q=79 exact-branch audit.
- A11/q79_exact_theorem (*derived exact*).
CRT q=79 theorem on the selected exact branch.

No imported row changes theorem ownership or promotes a neighboring claim: all local statements retain their stated hypotheses, domains, and limitations.

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